

PASSAGE

Passage republishes the Sparkplug namespace SparkBridge Host holds as a live OPC-UA server address space, so any OPC-UA client can browse and subscribe without polling the plant.

 FREE MODULE IGNITION 8.1.19+ AND 8.3 RUNS ON: HOST

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1. OVERVIEW

What Passage does

Passage republishes the Sparkplug namespace that SparkBridge Host holds as a live OPC-UA server address space. A non-Sparkplug client (a historian, a SCADA package, an MES, an analytics tool, a test rig) connects to Passage as an ordinary OPC-UA server and browses the same tag tree Host already built, without polling the plant. Passage is not a device driver and it does not write to the field; it exposes what Host already holds.

Passage ships as a free module included with any SparkBridge Edge or Host license.

- A SparkBridge Host gateway (Passage reads Host's tag tree).
- Ignition 8.1.19 or newer, or Ignition 8.3.
- A free TCP port for the OPC-UA endpoint (default 62544).
- An OPC-UA client to consume it: a historian, SCADA package, MES, analytics tool or test rig.

3. INSTALL

1. Download the SparkBridge release bundle for your Ignition line from [the download page](#).
2. In the gateway, go to Config > Modules > Install or Upgrade a Module, and upload Provider, then Host, then Edge, in that order.
3. Accept the certificate and license once; confirm all modules show Running. Passage is included, not a separate module upload.
4. Upgrade every installed module together, from the same release bundle, in the same maintenance window.

Passage is free. It is included with any SparkBridge Edge or Host license and ships with no license tag required of its own.

The suite key file, when present, lives at `<data>/sparkbridge-license.key` and is re-checked every 30 seconds, so installing or replacing it takes effect with no restart. From release 2.7.0, SparkBridge modules run on Ignition's two-hour module trial until a verified key names the product; Passage's own license state is not gated the same way since it carries no separate license tag.

5. CONFIGURE

Current Passage configuration is file/property-based. Set these as JVM system properties in the gateway startup configuration, then restart the gateway.

UNSBRIDGE.SERVER.ENABLED	Enable the OPC-UA server. Default true.
UNSBRIDGE.SERVER.PORT	OPC-UA bind port. Default 62544.
UNSBRIDGE.SERVER.BINDADDRESS	OPC-UA bind address. Default 127.0.0.1.
UNSBRIDGE.SERVER.HOSTNAME	Hostname advertised in OPC-UA endpoints. Default localhost.
UNSBRIDGE.SERVER.PATH	OPC-UA endpoint path. Default /unsbridge.
UNSBRIDGE.SERVER.SECURITYDIR	OPC-UA security directory. Default data/unsbridge/security.
UNSBRIDGE.SPIKE.ENABLED	Enable the synthetic proof node. Default false.
UNSBRIDGE.SERVER.ALLOWANONYMOUSUNENCRYPTEDREMOTE	Explicit acknowledgement for remote anonymous unencrypted access. Default false.
UNSBRIDGE.BIRTHCACHE.ENABLED	Persist and replay the last known BIRTH set. Default true.
UNSBRIDGE.BIRTHCACHE.FILE	Birth cache path. Default data/unsbridge-birthcache.txt.
UNSBRIDGE.BIRTHCACHE.MAXAGEHOURS	Maximum accepted cache age in hours. Default 168; 0 disables the age check.
UNSBRIDGE.BIRTHCACHE.SAVESECONDS	Cache save interval in seconds. Default 60; minimum 5.
UNSBRIDGE.BIRTHCACHE.DEVICEGRACESECONDS	Device grace period after an edge NBIRTH. Default 60; minimum 1.

`UNSBRIDGE.INGEST.ENABLED`

Enable Sparkplug MQTT ingest.
Default `false`.

`UNSBRIDGE.BROKER.URI`

MQTT broker URI. Default
`tcp://127.0.0.1:1883`.

`UNSBRIDGE.BROKER.USERNAME`

MQTT broker username. Default is
empty.

`UNSBRIDGE.BROKER.CLIENTID`

MQTT client id. Default
`sparkbridge-unsbridge`; use a
value beginning with `sparkbridge-`.

Store `unsbridge.status.token` and `unsbridge.broker.password` in the file
named by `unsbridge.secrets.file`, default `data/unsbridge-`
`secrets.properties`. Passage also supports the shared
`sparkbridge.broker.password` fallback for broker access.

Passage runs passively: it serves subscriptions and browse requests against Host's live tag tree. It polls nothing in the plant. On an edge NDEATH or DDEATH, the corresponding OPC-UA nodes go to bad quality within the configured device grace period. On gateway restart, the birth cache (when enabled) replays the last known BIRTH set so clients see a populated address space immediately rather than an empty tree while Host re-establishes its own sessions.

7. VERIFY

Connect a third-party OPC-UA client (for example UaExpert) to the configured endpoint, browse the address space, and subscribe to a value. Confirm the value matches the Sparkplug wire within one update, with the source timestamp preserved. Confirm a node goes to bad quality within the death window after an NDEATH/DDEATH.

SYMPTOM

CAUSE

`unsbridge.server.enabled` is false, or the port is blocked

`unsbridge.server.allowAnonymousUnencryptedRemote` is false (default)

Birth cache disabled or stale beyond
`unsbridge.birthcache.maxAgeHours`

`unsbridge.ingest.enabled` is false (default)

9. REFERENCE

- Default OPC-UA port: 62544.
- Default endpoint path: /unsbridge.
- Secrets file default: data/unsbridge-secrets.properties.
- Birth cache file default: data/unsbridge-birthcache.txt.